

Recitation 1

Maximum Likelihood Estimation

Solutions and Teaching Notes

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Pacing guide (75 minutes)

Time	Activity	Suggested use
0–25 min	MLE review	Setup, Poisson derivation, Gaussian derivation
25–37 min	Problem 1	7 min individual work, then discussion
37–50 min	Problem 2	7 min individual work, then discussion
50–66 min	Problem 3	8 min individual work, then discussion
66–75 min	Wrap-up or Problem 4	Use Problem 4 only if the class is ahead

1 Maximum likelihood estimation: the basic setup

Course notation. A tilde marks a random quantity: $\tilde{x}_1, \dots, \tilde{x}_n$. After observing these random variables, their realizations are $x_1, \dots, x_n$, and $X := \{x_1, \dots, x_n\}$ denotes the dataset. Thus, in this course, X is a dataset rather than a random variable. Model parameters and values fitted to the observed dataset are written without tildes or hats.

Suppose $\tilde{x}_1, \dots, \tilde{x}_n$ are i.i.d. random variables from a model indexed by a parameter $\theta \in \Theta$, and let $X = \{x_1, \dots, x_n\}$ be the observed dataset. The **likelihood** is

$$L(\theta; X) = \prod_{i=1}^n p_{\theta}(x_i) \quad \text{or} \quad L(\theta; X) = \prod_{i=1}^n f_{\theta}(x_i),$$

depending on whether the model is discrete or continuous. Once the data are observed, L is a function of θ ; it is not a pmf or pdf in θ .

Definition. The maximum likelihood (ML) estimate for the observed dataset X is

$$\theta_{\text{ML}} \in \arg \max_{\theta \in \Theta} L(\theta; X).$$

Because $\log$ is strictly increasing, maximizing L is equivalent to maximizing the log-likelihood

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log p_{\theta}(x_i) \quad \text{or} \quad \ell(\theta) = \sum_{i=1}^n \log f_{\theta}(x_i).$$

A reliable workflow is:

1. state the parameter space and the model assumptions;
2. write the likelihood, including any parameter-dependent support;
3. take logs and remove only terms that do not depend on the parameter;
4. find candidate maximizers and check the boundary or support;
5. report the estimate in the units of the original parameter.

2 Poisson MLE

Let $\tilde{x}_1, \dots, \tilde{x}_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$, where $\lambda \geq 0$. For the observed counts $X = \{x_1, \dots, x_n\}$,

$$L(\lambda) = \prod_{i=1}^n e^{-\lambda} \frac{\lambda^{x_i}}{x_i!},$$

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i \right) \log \lambda - \sum_{i=1}^n \log(x_i!).$$

If $\sum_i x_i > 0$, then

$$\ell'(\lambda) = -n + \frac{\sum_i x_i}{\lambda}, \quad \ell''(\lambda) = -\frac{\sum_i x_i}{\lambda^2} < 0.$$

Therefore

$$\lambda_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n x_i = m(X) = \bar{x}.$$

The estimate has a natural interpretation: the fitted expected count equals the sample average. If every observed count is zero, the likelihood decreases as λ increases, so the maximizer is the boundary value $\lambda_{\text{ML}} = 0$.

3 Gaussian MLE

Let $\tilde{x}_1, \dots, \tilde{x}_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, and let $X = \{x_1, \dots, x_n\}$ be the observed dataset. Write $v = \sigma^2$. The log-likelihood is

$$\ell(\mu, v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum_{i=1}^n (x_i - \mu)^2.$$

For fixed v ,

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{v} \sum_{i=1}^n (x_i - \mu) = 0 \implies \mu_{\text{ML}} = m(X) = \bar{x}.$$

The identity

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\mu - \bar{x})^2$$

also shows directly that $\mu = \bar{x}$ gives the global maximum for fixed v . Substitute $\mu_{\text{ML}} = \bar{x}$ and let $S = \sum_i (x_i - \bar{x})^2$. Then

$$\ell(\bar{x}, v) = C - \frac{n}{2} \log v - \frac{S}{2v}, \quad \frac{d\ell}{dv} = -\frac{n}{2v} + \frac{S}{2v^2}.$$

Thus, when $S > 0$,

$$\sigma_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \sigma_{\text{ML}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

MLE variance versus the usual sample variance. The Gaussian MLE divides by n . The unbiased sample variance divides by $n - 1$. Maximum likelihood targets the parameter value that best fits the observed data; it does not guarantee an unbiased estimator.

Degenerate data. If all observations are equal, then $S = 0$. With parameter space $v > 0$, the likelihood grows without bound as $v \downarrow 0$, so no MLE for v exists in that open parameter space. This is a useful reminder to check the boundary.

Practice problems

Problem 1: A two-point model

[7 minutes]

Suppose $\tilde{x}$ is a discrete random variable with

$$\mathbb{P}(\tilde{x} = 1) = \theta, \quad \mathbb{P}(\tilde{x} = 2) = 1 - \theta, \quad 0 \leq \theta \leq 1.$$

We draw n independent observations and find that n_1 of them equal 1 and n_2 equal 2, where $n_1 + n_2 = n$.

- (a) Write the likelihood function.
- (b) Find the maximum likelihood estimate of θ , including the cases $n_1 = 0$ or $n_2 = 0$.
- (c) Rewrite the MLE in terms of the sample mean $\bar{x}$.

Solution. The likelihood is

$$L(\theta) = \theta^{n_1} (1 - \theta)^{n_2}.$$

For $n_1, n_2 > 0$,

$$\ell'(\theta) = \frac{n_1}{\theta} - \frac{n_2}{1 - \theta} = 0 \implies \theta_{\text{ML}} = \frac{n_1}{n}.$$

Also $\ell''(\theta) < 0$ on $(0, 1)$. The same formula covers the boundaries: $n_1 = 0$ gives $\theta_{\text{ML}} = 0$, and $n_2 = 0$ gives $\theta_{\text{ML}} = 1$. Finally,

$$\bar{x} = \frac{n_1 + 2n_2}{n} = 2 - \frac{n_1}{n} \implies \theta_{\text{ML}} = 2 - \bar{x}.$$

Problem 2: Calls to a help desk

[7 minutes]

The numbers of calls received in eight comparable 15-minute periods are

$$3, 0, 2, 1, 4, 2, 0, 4.$$

Assume the counts are realizations of $\tilde{x}_1, \dots, \tilde{x}_8 \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$.

- (a) Find the MLE of λ .
- (b) Using the fitted model, estimate the probability that a future 15-minute period has at least three calls.
- (c) Give one modeling assumption that would make the calculation in part (b) questionable.

Solution. The total count is 16, so

$$\lambda_{\text{ML}} = \bar{x} = \frac{16}{8} = 2 \quad \text{calls per 15 minutes.}$$

If $\tilde{y} \sim \text{Poisson}(2)$ is the number of calls in a future period, then

$$\begin{aligned} \mathbb{P}_{\lambda_{\text{ML}}}(\tilde{y} \geq 3) &= 1 - \mathbb{P}_{\lambda_{\text{ML}}}(\tilde{y} = 0) - \mathbb{P}_{\lambda_{\text{ML}}}(\tilde{y} = 1) - \mathbb{P}_{\lambda_{\text{ML}}}(\tilde{y} = 2) \\ &= 1 - e^{-2} \left(1 + 2 + \frac{2^2}{2} \right) = 1 - 5e^{-2} \approx 0.323. \end{aligned}$$

Possible concerns include dependence across adjacent periods, different call rates at different times, or overdispersion relative to a Poisson model. Any one well-explained concern is sufficient.

Problem 3: Repeated measurements**[8 minutes]**

A device records the same reference quantity five times:

$$9.7, 10.1, 10.2, 9.9, 10.1.$$

Assume these values are realizations of $\tilde{x}_1, \dots, \tilde{x}_5 \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

- (a) Compute μ_{ML} .
- (b) Compute σ_{ML}^2 and σ_{ML} .
- (c) Compute the usual unbiased sample variance s^2 . Why does its denominator differ from the MLE denominator?
- (d) If 2 is added to every observation, how do the two MLEs change?

Solution. The sample mean is

$$\mu_{\text{ML}} = \bar{x} = 10.0.$$

The centered observations are $-0.3, 0.1, 0.2, -0.1, 0.1$, whose squared values sum to 0.16. Therefore

$$\sigma_{\text{ML}}^2 = \frac{0.16}{5} = 0.032, \quad \sigma_{\text{ML}} = \sqrt{0.032} \approx 0.179.$$

The unbiased sample variance is

$$s^2 = \frac{0.16}{4} = 0.04.$$

Estimating μ uses one degree of freedom, so division by $n - 1$ corrects the downward bias. Maximum likelihood instead maximizes the Gaussian likelihood and therefore divides by n . Adding 2 to every observation changes μ_{ML} from 10.0 to 12.0 and leaves σ_{ML}^2 unchanged.

Problem 4 (backup): True or false?

Optional if time allows. Suggested working time: 5–8 minutes.

For each statement, decide whether it is true or false and give a brief reason.

- (a) MLEs are always unbiased.
- (b) If $\tilde{\theta}_n \xrightarrow{p} \theta$, then $\tilde{\theta}_n^2 \xrightarrow{p} \theta^2$.
- (c) Every solution of the score equation $\ell'(\theta) = 0$ is an MLE.
- (d) A likelihood must satisfy $\int_{-\infty}^{\infty} L(\theta) d\theta = 1$.

Solution. The answers are false, true, false, false.

- (a) The Gaussian MLE of σ^2 is a counterexample: its expectation is $\frac{n-1}{n}\sigma^2$.
- (b) This follows from the continuous mapping theorem because $g(t) = t^2$ is continuous. The premise explicitly assumes consistency.
- (c) A zero derivative only identifies a stationary point. It may be a local minimum, and it can miss a maximizer on the boundary.
- (d) A likelihood is a function of the parameter with the data held fixed. It need not integrate to one over the parameter space. A normalized density in θ arises only after adding a prior and forming a posterior in a Bayesian analysis.

Takeaways

- For i.i.d. data, likelihoods multiply and log-likelihoods add.
- The Poisson MLE matches the sample mean count.
- The Gaussian MLE matches the sample mean and uses divisor n for the variance.
- A stationary point is not enough: check the parameter space, boundary, and parameter-dependent support.

Sources

The review follows the course handouts [Comparing Parametric and Nonparametric Models](#) and [Maximum Likelihood Estimation for Continuous Models](#). The notation follows Section 2.1 of the [Probability and Statistics for Data Science preprint](#). Problem 1 and Problem 4 are adapted from the 2025 Fall statistics recitation materials (Recitation 5).